

65. MIPI PHY

65.1 Overview

This MIPI PHY is a MIPI D-PHY module. The D-PHY module supports MIPI Alliance Specification Version 2.1 for D-PHY Specification.

Table 65.1 MIPI PHY specifications

Item		Specifications
D-PHY	Number of lanes	Up to 2 lanes
	Maximum rate	720 Mbps/lane
MIPI PLL circuit	Input clock source	MOSC
	Input pulse frequency division ratio	Selectable from 1/2/3/4
	Input clock frequency	8 MHz to 48 MHz
	Input clock frequency (after DPHYPLFCR.IDIV[1:0] bits	8 MHz to 24 MHz
	Frequency multiplication ratio	Selectable from 40 to 375 (after the decimal point: 0/0.33/0.50/0.66)
	VCO frequency	960 MHz to 3000 MHz
	Output Frequency Division Ratio Select	Selectable from 1/1, 1/2, 1/4, 1/8
	PLL output clock	PMUL[1:0] = 2'b00: 960 MHz to 1440 MHz PMUL[1:0] = 2'b01: 480 MHz to 1440 MHz PMUL[1:0] = 2'b10: 240 MHz to 750 MHz PMUL[1:0] = 2'b11: 120 MHz to 375 MHz

For the block diagram and I/O pins, see [Figure 62.1](#) and [Table 62.1](#).

65.2 Register Description

Access the following registers only in units of 32 bits.

65.2.1 DPHYREFCR : D-PHY Reference Clock Setting Register

Base address: MIPI_PHY0 = 0x4034_6C00
MIPI_PHY0 NS = 0x5034_6C00

Offset address: 0x000

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 0

[illegible]

Value after reset: 0 1 0 1 1 1 1

Bit	Symbol	Function	R/W
7:0	RFREQ[7:0]	Reference Clock Frequency Setting Set the frequency of the peripheral module clock (PCLKA^{*1}). PCLKA is used for the reference clock of the counter circuit which controls the internal timing of the D-PHY module. 0x27: 40 MHz^{*2} 0x28: 41 MHz 0x29: 42 MHz ⋮ ⋮ 0x7A: 123 MHz 0x7B: 124 MHz 0x7C: 125 MHz Others: Settings prohibited	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Remove digits after the decimal point. For example, PCLKA: 100.5 MHz → RFREQ[7:0] = 01100011b (100 MHz).

Note 2. The frequency of PCLKA must be 40 MHz or more to use the MIPI DSI function. Therefore, when using the MIPI DSI function, the operating power control mode of the system must also be set to high-speed-mode (OPCCR.OPCM[1:0] = 00b).

RFREQ[7:0] bits (Reference Clock Frequency Setting)

Set the RFREQ[7:0] bits to the frequency of the peripheral module clock (PCLKA)*1.

PCLKA is used for the reference clock of the counter circuit which controls the internal timing of the D-PHY module.

65.2.2 DPHYPLFCR : D-PHY PLL Frequency Control Register

Base address: MIPI_PHY0 = 0x4034_6C00
MIPI_PHY0_NS = 0x5034_6C00

Offset address: 0x004

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	NMUL[8:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	PMUL[1:0]		—	—	NFMUL[1:0]		—	—	—	—	—	—	IDIV[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	IDIV[1:0]	D-PHY PLL Input Frequency Division Ratio Select 0 0: 1 0 1: 1/2 1 0: 1/3 1 1: 1/4	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
9:8	NFMUL[1:0]	D-PHY PLL Frequency Multiplication Factor Select (Fractional Part) 0 0: 0.00 0 1: 0.33 1 0: 0.66 1 1: 0.50	R/W
11:10	—	These bits are read as 0. The write value should be 0.	R/W
13:12	PMUL[1:0]	D-PHY PLL Output Frequency Division Ratio Select 0 0: 1 0 1: 1/2 1 0: 1/4 1 1: 1/8	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W
24:16	NMUL[8:0]	D-PHY PLL Frequency Multiplication Factor Select (Integer Part) 0x27: 40 0x28: 41 0x29: 42 ⋮ 0x9E: 159 0x9F: 160 0xA0: 161 ⋮ 0x174: 373 0x175: 374 0x176: 375 Others: Setting prohibited	R/W
31:25	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

IDIV[1:0] bits (D-PHY PLL Input Frequency Division Ratio Select)

The IDIV[1:0] bits control the input division ratio I for D-PHY PLL.

NFMUL[1:0] bits (D-PHY PLL Frequency Multiplication Factor Select (Fractional Part))

The NFMUL[1:0] bits control the fractional part NF of the multiplication factor for D-PHY PLL.

PMUL[1:0] bits (D-PHY PLL Output Frequency Division Ratio Select)

The PMUL[1:0] bits control the output division ratio P for D-PHY PLL.

NMUL[8:0] bits (D-PHY PLL Frequency Multiplication Factor Select (Integer Part))

The NMUL[8:0] bits control the integer part N of the multiplication factor for D-PHY PLL.

The frequency of the PLL output clock (f_{DPHYPLL}) can be calculated by the following formula.

$$f_{\text{DPHYPLL}} [\text{MHz}] = f_{\text{MAIN}} [\text{MHz}] \times I \times (NF + N) \times P$$

Where, f_{MAIN} is the frequency of the main clock oscillator.

The D-PHY high-speed data transmission rate is determined by the following formula:

$$\text{Line rate [Mbps]} = f_{\text{DPHYPLL}} [\text{MHz}] / 2$$

Example:

When $f_{\text{MAIN}} = 24.576 \text{ MHz}$, IDIV[1:0] = 01b, NFMUL[1:0] = 11b, NMUL[8:0] = 0x63, PMUL[1:0] = 00b

$$f_{\text{DPHYPLL}} = 24.576 [\text{MHz}] \times 1/2 \times (0.50 + 100) \times 1/1 = 1235 [\text{MHz}]$$

$$\text{Line rate [Mbps]} = f_{\text{DPHYPLL}} / 2 = 1235 \text{ MHz} / 2 = 617.47 \text{ Mbps}$$

Note: The input frequency to the PLL should be 8 to 48 MHz.

Note: IDIV[1:0] must be set so that the frequency of the clock divided by IDIV[1:0] is within the range that is between 8 to 24 MHz.

Note: Set the frequency to be in the range of 960 to 3000 MHz after multiplication by NF + N.
(NFMUL[1:0] and NMUL[8:0] bit settings).

Note: The N parameter of NMUL[8:0] should be set between 40 and 375.

Note: The P parameter of PMUL[1:0] should be set to the following frequency range as a result of f_{DPHYPLL} calculation.

P = 1: 960 - 1440 MHz

P = 1/2: 480 - 1440 MHz

P = 1/4: 240 - 750 MHz

P = 1/8: 120 - 375 MHz

Note: DPHYPLFCR must be set while D-PHY PLL operation is stopped (DPHYPLOC.R.PLLSTP = 1).

65.2.3 DPHYPLOC.R : D-PHY PLL Operation Control Register

Base address: MIPI_PHY0 = 0x4034_6C00
MIPI_PHY0_NS = 0x5034_6C00

Offset address: 0x008

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PLLST P
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
0	PLLSTP	D-PHY PLL Operation Control 0: Operate the PLL 1: Stop the PLL	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

PLLSTP bit (D-PHY PLL Operation Control)

The PLLSTP bit controls the operation permission of the D-PHY PLL.

65.2.4 DPHYESCCR : D-PHY Escape Mode Clock Control Register

Base address: MIPI_PHY0 = 0x4034_6C00
MIPI_PHY0 NS = 0x5034_6C00

Offset address: 0x00C

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 0

[illegible][illegible]

Bit	Symbol	Function	R/W
4:0	ESCDIV[4:0]	Escape Mode Transfer Clock Division Ratio 0x00: $\times 1/1$ 0x01: $\times 1/2$ 0x02: $\times 1/3$ 0x03: $\times 1/4$: : 0x1D: $\times 1/30$ 0x1E: $\times 1/31$ 0x1F: $\times 1/32$	R/W
31:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

ESCDIV[4:0] bits (Escape Mode Transfer Clock Division Ratio)

The ESCDIV[4:0] bits control the division ratio for a low-power state clock which is used for escape mode transfer.

The frequency of the low-power state clock can be calculated by the following formula:

$$f_{LPCLK} = f_{DPHYPLL} \times (1/16) \times (\text{division ratio set by ESCDIV}[4:0])^{*1 *2 *3}$$

Note 1. These bits should be set so that the ratio of $T_{LTX(MASTER)}/T_{LTX(SLAVE)}$ between this device and the slave is in the range of 2/3 to 3/2.

$$T_{LPX(MASTER)} = 1 / f_{LPCLK}$$

$T_{LPX(SLAVE)}$ is the transmitted length of any low-power state period on peripheral.

Note 2. These bits should be set so that the frequency of the escape mode transfer clock (f_{LPX}) in this device is in the range of 2 to 17 MHz.

Note 3. These bits must be set while D-PHY PLL operation is stopped (DPHYPLOC.R.PLLSTP = 1).

65.2.5 DPHYPWRCR : D-PHY Power Supplying Control Register

Base address: MIPI_PHY0 = 0x4034_6C00
MIPI_PHY0_NS = 0x5034_6C00

Offset address: 0x010

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PWRS EN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PWRSN	D-PHY Power Supplying Control 0: Disable D-PHY LDO operation (stop supplying VDD_DPHY) 1: Enable D-PHY LDO operation (supply VDD_DPHY)	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

PWRSN bit (D-PHY Power Supplying Control)

The PWRSN bit controls the operation permission of the D-PHY LDO.

65.2.6 DPHYSFR : D-PHY Status Flag Register

Base address: MIPI_PHY0 = 0x4034_6C00
MIPI_PHY0_NS = 0x5034_6C00

Offset address: 0x01C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	PLLSF	—	—	—	—	—	—	—	PWRS F
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PWRSF	D-PHY LDO Power-on Status Flag 0: D-PHY LDO is stopped or starting up 1: D-PHY LDO startup is completed (VDD_DPHY is stable)	R
7:1	—	These bits are read as 0.	R
8	PLLSF	D-PHY PLL Oscillation Stabilization Flag 0: D-PHY PLL clock is stopped or is not yet stable 1: D-PHY PLL clock is stable	R
31:9	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

PWRSF flag (D-PHY LDO Power-on Status Flag)

The PWRSF flag indicates whether the operation of the D-PHY LDO has stabilized or not.

PLLSF flag (D-PHY PLL Oscillation Stabilization Flag)

The PLLSF flag indicates whether the oscillation of the D-PHY PLL has stabilized or not.

65.2.7 DPHYOCR : D-PHY Operation Control Register

Base address: MIPI_PHY0 = 0x4034_6C00
MIPI_PHY0 NS = 0x5034_6C00

Offset address: 0x020

[illegible]

Bit	Symbol	Function	R/W
0	DPHYEN	D-PHY Operation Control 0: Disable D-PHY operation 1: Enable D-PHY operation	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

DPHYEN bit (D-PHY Operation Control)

The DPHYEN bit controls the operation permission of the D-PHY.

65.2.8 DPHYTIM1 : D-PHY Timing Control Register 1

Base address: MIPI_PHY0 = 0x4034_6C00
MIPI_PHY0 NS = 0x5034_6C00

Offset address: 0x024

[illegible]

Bit	Symbol	Function	R/W
18:0	TINIT[18:0]	D-PHY T_INIT Parameter Setting Set the D-PHY TINIT parameter.	R/W
31:19	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

TINIT[18:0] bits (D-PHY T_INIT Parameter Setting)

The TINIT[18:0] bits specify the T_{INIT} parameter, which is the minimum duration of the TINIT state.

The time is calculated by the following formula:

$$T_{INIT} = (T_{INIT}[18:0] + 1) \times (\text{period of the peripheral module clock (PCLKA)})$$

For details, see [Table 65.2](#) and [Table 65.3](#).

65.2.9 DPHYTIM2 : D-PHY Timing Control Register 2

Base address: MIPI_PHY0 = 0x4034_6C00
MIPI_PHY0_NS = 0x5034_6C00

Offset address: 0x028

Bit position:	31	30	29	28	27	26	25	24	23					16	15					8	7					0
Bit field:	—	—	—	—	—	—	—	—	—	TCLKMISS[7:0]					TCLKSETT[7:0]					TCLKPREP[7:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	TCLKPREP[7:0]	D-PHY T_CLK_PREPARE Parameter Setting Set the D-PHY T_CLK-PREPARE parameter shown in Figure 65.1 .	R/W
15:8	TCLKSETT[7:0]	D-PHY T_CLK_SETTLE Parameter Setting Set the D-PHY T_CLK-SETTLE parameter.	R/W
23:16	TCLKMISS[7:0]	D-PHY T_CLK_MISS Parameter Setting Set the D-PHY T_CLK-MISS parameter.	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

TCLKPREP[7:0] bits (D-PHY T_CLK_PREPARE Parameter Setting)

The TCLKPREP[7:0] bits specify the T_CLK-PREPARE parameter, which is the duration of the LP-00 state (immediately before entry to the HS-0 state) in the clock lane. See [Figure 65.1](#).

The time is calculated by the following formula:

$$T_{\text{CLK-PREPARE}} = (\text{TCLKPREP}[7:0] + 1) \times (\text{period of the peripheral module clock (PCLKA)})$$

For details, see [Table 65.2](#) and [Table 65.3](#).

TCLKSETT[7:0] bits (D-PHY T_CLK_SETTLE Parameter Setting)

These bits specify the period over which a transition to the HS state is to be ignored after the T_CLK_PREPARE period has begun in the clock lane.

The time is calculated by the following formula:

$$T_{\text{CLK-SETTLE}} = (\text{TCLKSETT}[7:0] + 1) \times (\text{period of the peripheral module clock (PCLKA)})$$

For details, see [Table 65.3](#).

TCLKMISS[7:0] bits (D-PHY T_CLK_MISS Parameter Setting)

These bits specify the period from detection of the absence of a clock until disabling of the HS-RX in the clock lane.

The time is calculated by the following formula:

$$T_{\text{CLK-MISS}} = (\text{TCLKMISS}[7:0]) \times (\text{period of the peripheral module clock (PCLKA)})$$

For details, see [Table 65.3](#).

65.2.10 DPHYTIM3 : D-PHY Timing Control Register 3

Base address: MIPI_PHY0 = 0x4034_6C00
MIPI_PHY0_NS = 0x5034_6C00

Offset address: 0x02C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15											8	7											0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																												
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	THSETT[7:0]										THSPREP[7:0]																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Bit	Symbol	Function	R/W
7:0	THSPREP[7:0]	D-PHY T_THS_PREPARE Parameter Setting Set the D-PHY T _{HS-PREPARE} parameter shown in Figure 65.1 .	R/W
15:8	THSETT[7:0]	D-PHY T_THS_SETTLE Parameter Setting Set the D-PHY T _{HS-SETTLE} parameter.	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

THSPREP[7:0] bits (D-PHY T_THS_PREPARE Parameter Setting)

The THSPREP[7:0] bits specify the T_{HS-PREPARE} parameter, which is the duration of the LP-00 state (immediately before entry to the HS-0 state) in the data lane. See [Figure 65.1](#).

The time is calculated by the following formula:

$$T_{HS-PREPARE} = (THSPREP[7:0] + 1) \times (\text{period of the peripheral module clock (PCLKA)})$$

For details, see [Table 65.2](#) and [Table 65.3](#).

THSETT[7:0] bits (D-PHY T_THS_SETTLE Parameter Setting)

These bits specify the period over which a transition to the HS state is to be ignored after the T_{CLK-PREPARE} period has begun in the data lane.

The time is calculated by the following formula:

$$T_{HS-SETTLE} = (THSETT[7:0] + 1) \times (\text{period of the peripheral module clock (PCLKA)})$$

For details, see [Table 65.3](#).

65.2.11 DPHYTIM4 : D-PHY Timing Control Register 4

Base address: MIPI_PHY0 = 0x4034_6C00
MIPI_PHY0_NS = 0x5034_6C00

Offset address: 0x030

Bit position:	31	24	23	16	15	8	7	0																																
Bit field:	<table><tr><td colspan="8">TCLKTRL[7:0]</td><td colspan="8">TCLKPOST[7:0]</td><td colspan="8">TCLKPRE[7:0]</td><td colspan="8">TCLKZERO[7:0]</td></tr></table>								TCLKTRL[7:0]								TCLKPOST[7:0]								TCLKPRE[7:0]								TCLKZERO[7:0]							
TCLKTRL[7:0]								TCLKPOST[7:0]								TCLKPRE[7:0]								TCLKZERO[7:0]																
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0										

Bit	Symbol	Function	R/W
7:0	TCLKZERO[7:0]	D-PHY T_CLK_ZERO Parameter Setting Set the D-PHY T _{CLK-ZERO} parameter shown in Figure 65.1 .	R/W
15:8	TCLKPRE[7:0]	D-PHY T_TCLK_PRE Parameter Setting Set the D-PHY T _{CLK-PRE} parameter shown in Figure 65.1 .	R/W
23:16	TCLKPOST[7:0]	D-PHY T_TCLK_POST Parameter Setting Set the D-PHY T _{CLK-POST} parameter shown in Figure 65.1 .	R/W
31:24	TCLKTRL[7:0]	D-PHY T_TCLK_TRAIL Parameter Setting Set the D-PHY T _{CLK-TRAIL} parameter shown in Figure 65.1 .	R/W

Note: S-TYPE-3, P-TYPE-3

TCLKZERO[7:0] bits (D-PHY T_CLK_ZERO Parameter Setting)

The TCLKZERO[7:0] bits specify the T_{CLK-ZERO} parameter. For details, see [Figure 65.1](#).

The time is calculated by the following formula:

$$T_{CLK-ZERO} = (TCLKZERO[7:0] + 1) \times (\text{period of the peripheral module clock (PCLKA)})$$

For details, see [Table 65.2](#).

TCLKPRE[7:0] bits (D-PHY T_TCLK_PRE Parameter Setting)

The TCLKPRE[7:0] bits specify the T_{CLK-PRE} parameter. For details, see [Figure 65.1](#).

The time is calculated by the following formula:

$$T_{CLK-PRE} = (TCLKPRE[7:0] + 1) \times (\text{period of the peripheral module clock (PCLKA)})$$

For details, see [Table 65.2](#).

TCLKPOST[7:0] bits (D-PHY T_TCLK_POST Parameter Setting)

The TCLKPOST[7:0] bits specify the T_{CLK-POST} parameter. For details, see [Figure 65.1](#).

The time is calculated by the following formula:

$$T_{CLK-POST} = (TCLKPOST[7:0] + 1) \times (\text{period of the peripheral module clock (PCLKA)})$$

For details, see [Table 65.2](#).

TCLKTRL[7:0] bits (D-PHY T_TCLK_TRAIL Parameter Setting)

The TCLKTRL[7:0] bits specify the T_{CLK-TRAIL} parameter. For details, see [Figure 65.1](#).

The time is calculated by the following formula:

$$T_{CLK-TRAIL} = (TCLKTRL[7:0] + 1) \times (\text{period of the peripheral module clock (PCLKA)})$$

For details, see [Table 65.2](#).

65.2.12 DPHYTIM5 : D-PHY Timing Control Register 5

Base address: MIPI_PHY0 = 0x4034_6C00
MIPI_PHY0_NS = 0x5034_6C00

Offset address: 0x034

Bit position:	31	30	29	28	27	26	25	24	23		16	15		8	7		0
Bit field:	—	—	—	—	—	—	—	—	—	THSEXIT[7:0]			THSTRL[7:0]			THSZERO[7:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	THSZERO[7:0]	D-PHY T_THS_ZERO Parameter Setting Set the D-PHY T _{HS-ZERO} parameter shown in Figure 65.1 .	R/W
15:8	THSTRL[7:0]	D-PHY T_THS_TRAIL Parameter Setting Set the D-PHY T _{HS-TRAIL} parameter shown in Figure 65.1 .	R/W
23:16	THSEXIT[7:0]	D-PHY T_THS_EXIT Parameter Setting Set the D-PHY T _{HS-EXIT} parameter shown in Figure 65.1 .	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

THSZERO[7:0] bits (D-PHY T_THS_ZERO Parameter Setting)

The THSZERO[7:0] bits specify the T_{HS-ZERO} parameter. For details, see [Figure 65.1](#).

The time is calculated by the following formula:

$$T_{HS-ZERO} = (THSZERO[7:0] + 1) \times (\text{period of the peripheral module clock (PCLKA)})$$

For details, see [Table 65.2](#).

THSTRL[7:0] bits (D-PHY T_THS_TRAIL Parameter Setting)

The THSTRL[7:0] bits specify the T_{HS-TRAIL} parameter. For details, see [Figure 65.1](#).

The time is calculated by the following formula:

$$T_{HS-TRAIL} = (THSTRL[7:0] + 1) \times (\text{period of the peripheral module clock (PCLKA)})$$

For details, see [Table 65.2](#).

THSEXIT[7:0] bits (D-PHY T_THS_EXIT Parameter Setting)

The THSEXIT[7:0] bits specify the T_{HS-EXIT} parameter. For details, see [Figure 65.1](#).

The time is calculated by the following formula:

$$T_{HS-EXIT} = (THSEXIT[7:0] + 1) \times (\text{period of the peripheral module clock (PCLKA)})$$

For details, see [Table 65.2](#).

65.2.13 DPHYTIM6 : D-PHY Timing Control Register 6

Base address: MIPI_PHY0 = 0x4034_6C00
MIPI_PHY0 NS = 0x5034_6C00

Offset address: 0x038

[illegible]

Bit	Symbol	Function	R/W
7:0	TLPX[7:0]	D-PHY T_TLPX Parameter Setting Set the D-PHY T_LPX parameter shown in Figure 65.1 .	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

TLPX[7:0] bits (D-PHY T_TLPX Parameter Setting)

The THSZERO[7:0] bits specify the T_{TPX} parameter. For details, see [Figure 65.1](#).

The time is calculated by the following formula:

$$T_{IPX} = (TLPX[7:0] + 1) \times (\text{period of the peripheral module clock (PCLKA)})$$

For details, see [Table 65.2](#).

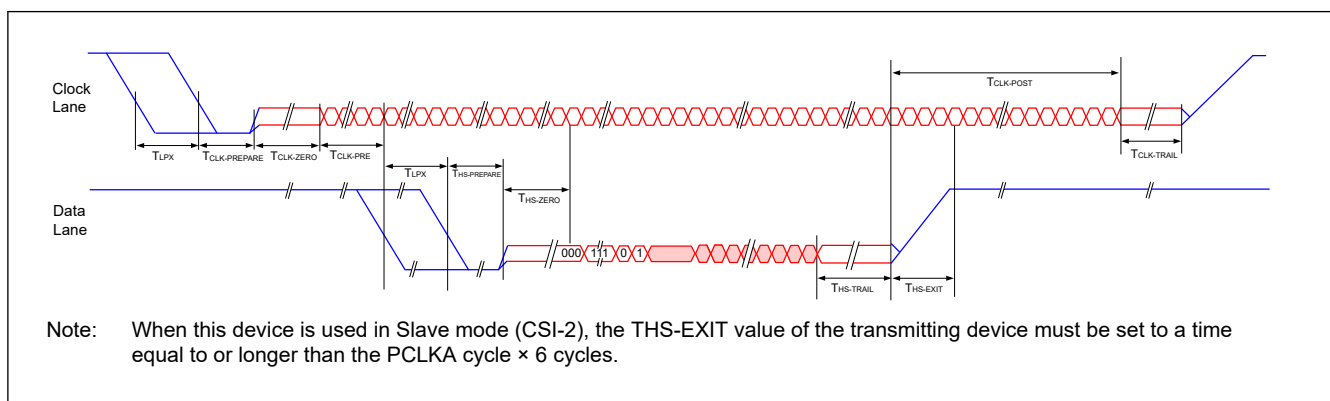


Figure 65.1 High-speed data transmission in normal mode

Table 65.2 D-PHY timing setting DSI mode (1 of 7)[illegible]

Table 65.2 D-PHY timing setting DSI mode (2 of 7)

No.	Bit name	PCLKA 125 MHz					
		80 Mbps to 100 Mbps	100 Mbps to 150 Mbps	150 Mbps to 250 Mbps	250 Mbps to 400 Mbps	400 Mbps to 600 Mbps	600 Mbps to 720 Mbps
2	TCLKPREP[7:0]	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)
3	THSPREP[7:0]	15 (0xF)	13 (0xD)	11 (0xB)	10 (0xA)	9 (0x9)	8 (0x8)
4	TCLKZERO[7:0]	33 (0x21)	33 (0x21)	33 (0x21)	33 (0x21)	33 (0x21)	33 (0x21)
5	TCLKPRE[7:0]	13 (0xD)	13 (0xD)	13 (0xD)	4 (0x4)	4 (0x4)	4 (0x4)
6	TCLKPOST[7:0]	88 (0x58)	88 (0x58)	58 (0x3A)	58 (0x3A)	35 (0x23)	35 (0x23)
7	TCLKTRL[7:0]	10 (0xA)	10 (0xA)	8 (0x8)	8 (0x8)	6 (0x6)	5 (0x5)
8	THSZERO[7:0]	15 (0xF)	15 (0xF)	15 (0xF)	15 (0xF)	15 (0xF)	15 (0xF)
9	THSTRL[7:0]	22 (0x16)	17 (0x11)	12 (0xC)	9 (0x9)	8 (0x8)	7 (0x7)
10	THSEXIT[7:0]	14 (0xE)	14 (0xE)	14 (0xE)	14 (0xE)	14 (0xE)	14 (0xE)
11	TLPX[7:0]	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)

Table 65.2 D-PHY timing setting DSI mode (3 of 7)

No.	Bit name	PCLKA 120 MHz					
		80 Mbps to 100 Mbps	100 Mbps to 150 Mbps	150 Mbps to 250 Mbps	250 Mbps to 400 Mbps	400 Mbps to 600 Mbps	600 Mbps to 720 Mbps
1	TINIT[18:0]	72001 (0x11941)	72001 (0x11941)	72001 (0x11941)	72001 (0x11941)	72001 (0x11941)	72001 (0x11941)
2	TCLKPREP[7:0]	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)
3	THSPREP[7:0]	14 (0xE)	12 (0xC)	10 (0xA)	9 (0x9)	9 (0x9)	8 (0x8)
4	TCLKZERO[7:0]	32 (0x20)	32 (0x20)	32 (0x20)	32 (0x20)	32 (0x20)	32 (0x20)
5	TCLKPRE[7:0]	13 (0xD)	13 (0xD)	13 (0xD)	4 (0x4)	4 (0x4)	4 (0x4)
6	TCLKPOST[7:0]	88 (0x58)	88 (0x58)	58 (0x3A)	58 (0x3A)	35 (0x23)	35 (0x23)
7	TCLKTRL[7:0]	10 (0xA)	10 (0xA)	8 (0x8)	7 (0x7)	6 (0x6)	5 (0x5)
8	THSZERO[7:0]	15 (0xF)	15 (0xF)	15 (0xF)	15 (0xF)	15 (0xF)	15 (0xF)
9	THSTRL[7:0]	21 (0x15)	16 (0x10)	11 (0xB)	8 (0x8)	7 (0x7)	6 (0x6)
10	THSEXIT[7:0]	13 (0xD)	13 (0xD)	13 (0xD)	13 (0xD)	13 (0xD)	13 (0xD)
11	TLPX[7:0]	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)

Table 65.2 D-PHY timing setting DSI mode (4 of 7)

No.	Bit name	PCLKA 100 MHz					
		80 Mbps to 100 Mbps	100 Mbps to 150 Mbps	150 Mbps to 250 Mbps	250 Mbps to 400 Mbps	400 Mbps to 600 Mbps	600 Mbps to 720 Mbps
1	TINIT[18:0]	60001 (0xEA61)	60001 (0xEA61)	60001 (0xEA61)	60001 (0xEA61)	60001 (0xEA61)	60001 (0xEA61)
2	TCLKPREP[7:0]	6 (0x6)	6 (0x6)	6 (0x6)	6 (0x6)	6 (0x6)	6 (0x6)
3	THSPREP[7:0]	12 (0xC)	10 (0xA)	8 (0x8)	7 (0x7)	7 (0x7)	7 (0x7)
4	TCLKZERO[7:0]	28 (0x1C)	28 (0x1C)	28 (0x1C)	28 (0x1C)	28 (0x1C)	28 (0x1C)
5	TCLKPRE[7:0]	12 (0xC)	12 (0xC)	12 (0xC)	4 (0x4)	4 (0x4)	4 (0x4)
6	TCLKPOST[7:0]	70 (0x46)	70 (0x46)	50 (0x32)	34 (0x22)	34 (0x22)	34 (0x22)
7	TCLKTRL[7:0]	10 (0xA)	7 (0x7)	6 (0x6)	4 (0x4)	4 (0x4)	3 (0x3)
8	THSZERO[7:0]	12 (0xC)	12 (0xC)	12 (0xC)	12 (0xC)	12 (0xC)	12 (0xC)
9	THSTRL[7:0]	16 (0x10)	11 (0xB)	8 (0x8)	6 (0x6)	5 (0x5)	5 (0x5)
10	THSEXIT[7:0]	12 (0xC)	12 (0xC)	12 (0xC)	12 (0xC)	12 (0xC)	12 (0xC)
11	TLPX[7:0]	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)

Table 65.2 D-PHY timing setting DSI mode (5 of 7)

No.	Bit name	PCLKA 80 MHz					
		80 Mbps to 100 Mbps	100 Mbps to 150 Mbps	150 Mbps to 250 Mbps	250 Mbps to 400 Mbps	400 Mbps to 600 Mbps	600 Mbps to 720 Mbps
1	TINIT[18:0]	48001 (0xBB81)	48001 (0xBB81)	48001 (0xBB81)	48001 (0xBB81)	48001 (0xBB81)	48001 (0xBB81)
2	TCLKPREP[7:0]	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)
3	THSPREP[7:0]	9 (0x9)	8 (0x8)	7 (0x7)	6 (0x6)	5 (0x5)	5 (0x5)
4	TCLKZERO[7:0]	23 (0x17)	23 (0x17)	23 (0x17)	23 (0x17)	23 (0x17)	23 (0x17)
5	TCLKPRE[7:0]	10 (0xA)	10 (0xA)	4 (0x4)	4 (0x4)	4 (0x4)	4 (0x4)
6	TCLKPOST[7:0]	58 (0x3A)	46 (0x2E)	46 (0x2E)	28 (0x1C)	28 (0x1C)	28 (0x1C)
7	TCLKTRL[7:0]	7 (0x7)	5 (0x5)	4 (0x4)	3 (0x3)	2 (0x2)	2 (0x2)
8	THSZERO[7:0]	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)
9	THSTRL[7:0]	12 (0xC)	8 (0x8)	6 (0x6)	4 (0x4)	4 (0x4)	4 (0x4)
10	THSEXIT[7:0]	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)
11	TLPX[7:0]	7 (0x7)	7 (0x7)	7 (0x7)	7 (0x7)	7 (0x7)	7 (0x7)

Table 65.2 D-PHY timing setting DSI mode (6 of 7)

No	Bit name	PCLKA 75 MHz					
		80 Mbps to 100 Mbps	100 Mbps to 150 Mbps	150 Mbps to 250 Mbps	250 Mbps to 400 Mbps	400 Mbps to 600 Mbps	600 Mbps to 720 Mbps
1	TINIT[18:0]	45001 (0xAFC9)	45001 (0xAFC9)	45001 (0xAFC9)	45001 (0xAFC9)	45001 (0xAFC9)	45001 (0xAFC9)
2	TCLKPREP[7:0]	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)
3	THSPREP[7:0]	8 (0x8)	7 (0x7)	6 (0x6)	5 (0x5)	5 (0x5)	5 (0x5)
4	TCLKZERO[7:0]	20 (0x14)	20 (0x14)	20 (0x14)	20 (0x14)	20 (0x14)	20 (0x14)
5	TCLKPRE[7:0]	8 (0x8)	8 (0x8)	3 (0x3)	3 (0x3)	3 (0x3)	3 (0x3)
6	TCLKPOST[7:0]	55 (0x37)	43 (0x2B)	43 (0x2B)	25 (0x19)	25 (0x19)	25 (0x19)
7	TCLKTRL[7:0]	6 (0x6)	4 (0x4)	3 (0x3)	2 (0x2)	2 (0x2)	1 (0x1)
8	THSZERO[7:0]	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)
9	THSTRL[7:0]	11 (0xB)	8 (0x8)	5 (0x5)	4 (0x4)	4 (0x4)	4 (0x4)
10	THSEXIT[7:0]	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)
11	TLPX[7:0]	7 (0x7)	7 (0x7)	7 (0x7)	7 (0x7)	7 (0x7)	7 (0x7)

Table 65.2 D-PHY timing setting DSI mode (7 of 7)

No.	Bit name	PCLKA 40 MHz							
		80 Mbps to 100 Mbps	100 Mbps to 115 Mbps	115 Mbps to 125 Mbps	125 Mbps to 150 Mbps	150 Mbps to 250 Mbps	250 Mbps to 400 Mbps	400 Mbps to 600 Mbps	600 Mbps to 720 Mbps
1	TINIT[18:0]	24001 (0x5DC1)	24001 (0x5DC1)	24001 (0x5DC1)	24001 (0x5DC1)	24001 (0x5DC1)	24001 (0x5DC1)	24001 (0x5DC1)	24001 (0x5DC1)
2	TCLKPREP[7:0]	2 (0x2)	2 (0x2)	2 (0x2)	2 (0x2)	2 (0x2)	2 (0x2)	2 (0x2)	2 (0x2)
3	THSPREP[7:0]	4 (0x4)	4 (0x4)	3 (0x3)	3 (0x3)	3 (0x3)	2 (0x2)	2 (0x2)	2 (0x2)
4	TCLKZERO[7:0]	12 (0xC)	12 (0xC)	12 (0xC)	12 (0xC)	12 (0xC)	12 (0xC)	12 (0xC)	12 (0xC)
5	TCLKPRE[7:0]	2 (0x2)	2 (0x2)	2 (0x2)	2 (0x2)	0 (0x0)	0 (0x0)	0 (0x0)	0 (0x0)
6	TCLKPOST[7:0]	32 (0x20)	26 (0x1A)	26 (0x1A)	26 (0x1A)	26 (0x1A)	12 (0xC)	12 (0xC)	12 (0xC)
7	TCLKTRL[7:0]	1 (0x1)	1 (0x1)	1 (0x1)	1 (0x1)	0 (0x0)	0 (0x0)	0 (0x0)	0 (0x0)
8	THSZERO[7:0]	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)
9	THSTRL[7:0]	3 (0x3)	2 (0x2)	2 (0x2)	2 (0x2)	2 (0x2)	2 (0x2)	2 (0x2)	2 (0x2)
10	THSEXIT[7:0]	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)
11	TLPX[7:0]	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)	5 (0x5)

Table 65.3 D-PHY timing setting CSI mode (1 of 7)

No.	Bit name	PCLKA 125 MHz							
		80 Mbps to 90 Mbps	90 Mbps to 100 Mbps	100 Mbps to 130 Mbps	130 Mbps to 200 Mbps	200 Mbps to 300 Mbps	300 Mbps to 400 Mbps	400 Mbps to 700 Mbps	700 Mbps to 720 Mbps
1	TINIT[18:0]	75001 (0x124F9)	75001 (0x124F9)	75001 (0x124F9)	75001 (0x124F9)	75001 (0x124F9)	75001 (0x124F9)	75001 (0x124F9)	75001 (0x124F9)
2	TCLKMISS[7:0]	3 (0x3)	3 (0x3)	3 (0x3)	3 (0x3)	3 (0x3)	3 (0x3)	3 (0x3)	3 (0x3)
3	TCLKSETT[7:0]	25 (0x19)	25 (0x19)	25 (0x19)	25 (0x19)	25 (0x19)	25 (0x19)	25 (0x19)	25 (0x19)
4	THSETT[7:0]	18 (0x12)	17 (0x11)	16 (0x10)	15 (0xF)	14 (0xE)	13 (0xD)	12 (0xC)	11 (0xB)
5	TCLKPREP[7:0]	17 (0x11)	17 (0x11)	17 (0x11)	17 (0x11)	17 (0x11)	17 (0x11)	17 (0x11)	17 (0x11)
6	THSPREP[7:0]	18 (0x12)	17 (0x11)	16 (0x10)	15 (0xF)	13 (0xD)	12 (0xC)	11 (0xB)	10 (0xA)

Table 65.3 D-PHY timing setting CSI mode (2 of 7)

No.	Bit name	PCLKA 120 MHz							
		80 Mbps to 90 Mbps	90 Mbps to 100 Mbps	100 Mbps to 130 Mbps	130 Mbps to 200 Mbps	200 Mbps to 300 Mbps	300 Mbps to 400 Mbps	400 Mbps to 700 Mbps	700 Mbps to 720 Mbps
1	TINIT[18:0]	72001 (0x11941)	72001 (0x11941)	72001 (0x11941)	72001 (0x11941)	72001 (0x11941)	72001 (0x11941)	72001 (0x11941)	72001 (0x11941)
2	TCLKMISS[7:0]	3 (0x3)	3 (0x3)	3 (0x3)	3 (0x3)	3 (0x3)	3 (0x3)	3 (0x3)	3 (0x3)
3	TCLKSETT[7:0]	24 (0x18)	24 (0x18)	24 (0x18)	24 (0x18)	24 (0x18)	24 (0x18)	24 (0x18)	24 (0x18)
4	THSETT[7:0]	17 (0x11)	16 (0x10)	15 (0xF)	14 (0xE)	13 (0xD)	12 (0xC)	11 (0xB)	10 (0xA)
5	TCLKPREP[7:0]	16 (0x10)	16 (0x10)	16 (0x10)	16 (0x10)	16 (0x10)	16 (0x10)	16 (0x10)	16 (0x10)
6	THSPREP[7:0]	17 (0x11)	16 (0x10)	15 (0xF)	14 (0xE)	12 (0xC)	11 (0xB)	10 (0xA)	9 (0x9)

Table 65.3 D-PHY timing setting CSI mode (3 of 7)

No.	Bit name	PCLKA 100 MHz							
		80 Mbps to 90 Mbps	90 Mbps to 100 Mbps	100 Mbps to 130 Mbps	130 Mbps to 200 Mbps	200 Mbps to 300 Mbps	300 Mbps to 400 Mbps	400 Mbps to 700 Mbps	700 Mbps to 720 Mbps
1	TINIT[18:0]	60001 (0xEA61)	60001 (0xEA61)	60001 (0xEA61)	60001 (0xEA61)	60001 (0xEA61)	60001 (0xEA61)	60001 (0xEA61)	60001 (0xEA61)
2	TCLKMISS[7:0]	2 (0x2)	2 (0x2)	2 (0x2)	2 (0x2)	2 (0x2)	2 (0x2)	2 (0x2)	2 (0x2)
3	TCLKSETT[7:0]	19 (0x13)	19 (0x13)	19 (0x13)	19 (0x13)	19 (0x13)	19 (0x13)	19 (0x13)	19 (0x13)
4	THSETT[7:0]	14 (0xE)	13 (0xD)	12 (0xC)	11 (0xB)	10 (0xA)	9 (0x9)	8 (0x8)	8 (0x8)
5	TCLKPREP[7:0]	13 (0xD)	13 (0xD)	13 (0xD)	13 (0xD)	13 (0xD)	13 (0xD)	13 (0xD)	13 (0xD)
6	THSPREP[7:0]	14 (0xE)	13 (0xD)	12 (0xC)	11 (0xB)	10 (0xA)	9 (0x9)	8 (0x8)	7 (0x7)

Table 65.3 D-PHY timing setting CSI mode (4 of 7)

No.	Bit name	PCLKA 80 MHz							
		80 Mbps to 90 Mbps	90 Mbps to 100 Mbps	100 Mbps to 130 Mbps	130 Mbps to 200 Mbps	200 Mbps to 300 Mbps	300 Mbps to 400 Mbps	400 Mbps to 700 Mbps	700 Mbps to 720 Mbps
1	TINIT[18:0]	48001 (0xBB81)	48001 (0xBB81)	48001 (0xBB81)	48001 (0xBB81)	48001 (0xBB81)	48001 (0xBB81)	48001 (0xBB81)	48001 (0xBB81)
2	TCLKMISS[7:0]	1 (0x1)	1 (0x1)	1 (0x1)	1 (0x1)	1 (0x1)	1 (0x1)	1 (0x1)	1 (0x1)
3	TCLKSETT[7:0]	15 (0xF)	15 (0xF)	15 (0xF)	15 (0xF)	15 (0xF)	15 (0xF)	15 (0xF)	15 (0xF)
4	THSETT[7:0]	10 (0xA)	10 (0xA)	9 (0x9)	8 (0x8)	7 (0x7)	7 (0x7)	6 (0x6)	6 (0x6)

Table 65.3 D-PHY timing setting CSI mode (5 of 7)

No.	Bit name	PCLKA 80 MHz							
		80 Mbps to 90 Mbps	90 Mbps to 100 Mbps	100 Mbps to 130 Mbps	130 Mbps to 200 Mbps	200 Mbps to 300 Mbps	300 Mbps to 400 Mbps	400 Mbps to 700 Mbps	700 Mbps to 720 Mbps
5	TCLKPREP[7:0]	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)
6	THSPREP[7:0]	10 (0xA)	10 (0xA)	9 (0x9)	8 (0x8)	7 (0x7)	7 (0x7)	6 (0x6)	5 (0x5)

Table 65.3 D-PHY timing setting CSI mode (6 of 7)

No.	Bit name	PCLKA 75 MHz							
		80 Mbps to 90 Mbps	90 Mbps to 100 Mbps	100 Mbps to 130 Mbps	130 Mbps to 200 Mbps	200 Mbps to 300 Mbps	300 Mbps to 400 Mbps	400 Mbps to 700 Mbps	700 Mbps to 720 Mbps
1	TINIT[18:0]	45001 (0xAFC9)	45001 (0xAFC9)	45001 (0xAFC9)	45001 (0xAFC9)	45001 (0xAFC9)	45001 (0xAFC9)	45001 (0xAFC9)	45001 (0xAFC9)
2	TCLKMISS[7:0]	1 (0x1)	1 (0x1)	1 (0x1)	1 (0x1)	1 (0x1)	1 (0x1)	1 (0x1)	1 (0x1)
3	TCLKSETT[7:0]	14 (0xE)	14 (0xE)	14 (0xE)	14 (0xE)	14 (0xE)	14 (0xE)	14 (0xE)	14 (0xE)
4	THSETT[7:0]	9 (0x9)	9 (0x9)	8 (0x8)	7 (0x7)	6 (0x6)	6 (0x6)	6 (0x6)	6 (0x6)
5	TCLKPREP[7:0]	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)	10 (0xA)
6	THSPREP[7:0]	9 (0x9)	9 (0x9)	8 (0x8)	7 (0x7)	6 (0x6)	6 (0x6)	5 (0x5)	5 (0x5)

Table 65.3 D-PHY timing setting CSI mode (7 of 7)

No.	Bit name	PCLKA 50 MHz							
		80 Mbps to 90 Mbps	90 Mbps to 100 Mbps	100 Mbps to 130 Mbps	130 Mbps to 200 Mbps	200 Mbps to 300 Mbps	300 Mbps to 400 Mbps	400 Mbps to 700 Mbps	700 Mbps to 720 Mbps
1	TINIT[18:0]	30001 (0x7531)	30001 (0x7531)	30001 (0x7531)	30001 (0x7531)	30001 (0x7531)	30001 (0x7531)	30001 (0x7531)	30001 (0x7531)
2	TCLKMISS[7:0]	1 (0x1)	1 (0x1)	1 (0x1)	1 (0x1)	1 (0x1)	1 (0x1)	1 (0x1)	1 (0x1)
3	TCLKSETT[7:0]	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)	8 (0x8)
4	THSETT[7:0]	5 (0x5)	5 (0x5)	4 (0x4)	4 (0x4)	3 (0x3)	3 (0x3)	3 (0x3)	2 (0x2)
5	TCLKPREP[7:0]	6 (0x6)	6 (0x6)	6 (0x6)	6 (0x6)	6 (0x6)	6 (0x6)	6 (0x6)	6 (0x6)
6	THSPREP[7:0]	5 (0x5)	5 (0x5)	4 (0x4)	4 (0x4)	3 (0x3)	3 (0x3)	3 (0x3)	2 (0x2)

65.2.14 DPHYMDC : D-PHY Mode Control Register

Base address: MIPI_PHY0 = 0x4034_6C00
 MIPI_PHY0_NS = 0x5034_6C00

Offset address: 0x048

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MAST EREN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MASTEREN	D-PHY Master/Slave Select 0: Slave Mode (CSI-2 Version 3.0, D-PHY v2.1 CIL_SCNN,CIL_SFEN) 1: Master Mode (DSI-2 Version 1.1, D-PHY v2.1 CIL_MCNN,CIL_MFAA)	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

65.3 Operation

65.3.1 D-PHY Startup Procedure

To use the MIPI DSI or MIPI CSI function, D-PHY must first be activated. The D-PHY has a dedicated regulator (D-PHY LDO) and PLL (D-PHY PLL), and these regulators are activated when the D-PHY starts up.

The startup procedure for the D-PHY is as follows:

Before executing the following flow, the basic register settings of GLCDC must be completed. Especially, SYSCNT_PANEL_CLK.CLKEN must be set to 1 to supply the panel clock to the DSI host as the video clock.^{*1}

1. Confirm that Graphics power domain is supplied with power and PCLKA is supplied to the MIPI DSI or MIPI CSI subsystem.
2. Set DPHYMDC.MASTEREN (set to 1 for Master mode (DSI setting), set to 0 for Slave mode (CSI setting)).
3. Set the DPHYREFCR.RFREQ[7:0] bits for the frequency of PCLKA.
4. Set the DPHYPWRCR.PWRSEN bit to 1 to start operating the D-PHY LDO (for supplying VDD_DPHY).
5. Wait for the startup of VDD_DPHY while the DPHYSFR.PWRSF bit is 0.
6. Set the DPHYPLFCR register to determine the frequency of the D-PHY PLL clock^{*1}.
7. Set the DPHYESCCR.ESCDIV[4:0] bits to determine the frequency of the D-PHY escape mode clock^{*1}.
8. Clear the DPHYPLOC.R.PLLSTP bit to 0 to enable PLL operation^{*1}.
9. Wait for the oscillation stabilization while the DPHYSFR.PLLSF bit is 0^{*1}.
10. Set the following registers for the D-PHY timing parameters.
(DPHYTIM1, DPHYTIM2, DPHYTIM3, DPHYTIM4, DPHYTIM5, DPHYTIM6).
11. Set the DPHYOCR.DPHYEN bit to 1 to enable D-PHY operation.

Note 1. No setting is required for Slave mode (CSI setting).

65.3.2 D-PHY Stop Procedure

To stop the D-PHY, use the following steps.

Note: It is not possible to operate the D-PHY LDO in standby mode. Therefore, before going to standby mode, it is necessary to disable the D-PHY LDO operation.

1. Clear the DPHYOCR.DPHYEN bit to 0 to disable the D-PHY.
2. Set the DPHYPLOC.R.PLLSTP bit to 1 to stop the PLL operation.
3. Clear the DPHYPWRCR.PWRSEN bit to 0 to disable the D-PHY LDO (stop VDD_DPHY supply).

65.4 Usage Notes

65.4.1 Power Gating Control or Software Standby Mode

Before transitioning to graphics power domain off or Software Standby mode, stop the operation as described in [section 65.3.2. D-PHY Stop Procedure](#).

Then after returning from graphics power domain off or Software Standby mode, resume operation as described in [section 65.3.1. D-PHY Startup Procedure](#).

65.4.2 Module-Stop Function Setting

MIPI PHY operation can be disabled or enabled using Module Stop Control Register C (MSTPCRC). The MIPI PHY module is initially stopped after reset. Releasing the module-stop state enables access to the registers.